UNIT-4:

MATRIX DECOMPOSITION

Diagonalization:

Let A be a square matrix of order n.
Let $\lambda_1, \lambda_2, \dots \lambda_n$ be its eigen values and

$$x_1 = \begin{bmatrix} x_{11} \\ x_{12} \\ \vdots \\ x_{1n} \end{bmatrix} \quad X_2 = \begin{bmatrix} x_{21} \\ x_{22} \\ \vdots \\ x_{2n} \end{bmatrix}, \quad X_n = \begin{bmatrix} x_{n1} \\ x_{n2} \\ \vdots \\ x_{nn} \end{bmatrix}$$

be corresponding eigen vectors.
Let B be the square matrix whose columns are $X_1, X_2, \dots X_n$ then

$$P^{-1}AP = D$$

where $D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & & & \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}$

Thus if a square matrix A of order n has n eigen vectors then there exists a non-singular square matrix P of order n such that $P^{-1}AP$ is a diagonal matrix D.

NOTE:

1. The matrix P is called the diagonalizing matrix or the modal matrix of A & the matrix D is called the spectral matrix of A.
2. $A^k = PD^kP^{-1}$

Q.) Diagonalize the following matrices:

1. $\begin{bmatrix} -1 & 2 \\ 2 & -1 \end{bmatrix}$ & find A^6

Characteristic eq:

$|A-\lambda I|=0$

$\Rightarrow (-1-\lambda)(-1-\lambda)-4=0$

$1+\lambda+\lambda+\lambda^2-4=0$

$\lambda^2+2\lambda-3=0$

$\underline{\lambda=1,-3}$ are eigen values.

$\therefore D=\begin{bmatrix}1 & 0\\ 0 & -3\end{bmatrix}$

Now $[A-\lambda I][X]=[0]$

$\Rightarrow \left.\begin{aligned}(-1-\lambda)x+2y&=0\\ 2x+(-1-\lambda)y&=0\end{aligned}\right\}*$

Case 1: $\lambda=1$

$* \Rightarrow -2x+2y=0$

$2x-2y=0$

$\Rightarrow \underline{2x=2y}$

$x=y=k=1$ (say)

$X_1=\begin{bmatrix}1\\1\end{bmatrix}$

Case 2: $\lambda=-3$

$* \Rightarrow 2x+2y=0$

$2x+2y=0$

$\Rightarrow x=-y=k$

let $x=1$, $y=-1$

$X_2=\begin{bmatrix}1\\-1\end{bmatrix}$

$\Rightarrow \underline{P=\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}}$

Now, $A^6=PD^6P^{-1}$

$\underset{2\times 2}{\text{adj}}$ { Exchange diagonal elements & change sign of remaining

$$P^{-1} = \frac{1}{|P|} \cdot \text{adj}\, P$$

$$= \frac{1}{-2}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$

Note that $P^{-1}AP = D$, where P & D are as given above.

Thus given matrix A has been diagonalized.

$$A^6 = PD^6P^{-1}$$

$$= \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1^6 & 0 \\ 0 & (-3)^6 \end{bmatrix}\left(\frac{-1}{2}\right)\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \frac{-1}{2}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & +729 \end{bmatrix}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \frac{-1}{2}\begin{bmatrix} 1 & +729 \\ 1 & -729 \end{bmatrix}\begin{bmatrix} -1 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \frac{-1}{2}\begin{bmatrix} -730 & -728 \\ 728 & -730 \end{bmatrix}$$

$$= \begin{bmatrix} -365 & -365 \\ -364 & +365 \end{bmatrix}$$

2. $\begin{bmatrix} 3 & 1 \\ 2 & 2 \end{bmatrix}$

Characteristic eqn: $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 1 \\ 2 & 2-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)(2-\lambda) - 2 = 0$$

$$6 - 3\lambda - 2\lambda + \lambda^2 - 2 = 0$$

$$\lambda^2 - 5\lambda + 4 = 0$$

$\lambda = 4, 1$ are eigen values $\therefore D = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}$

Now $[A-\lambda I][X]=[0]$

$$\left.\begin{aligned}(3-\lambda)x+y&=0\\ 2x+(2-\lambda)y&=0\end{aligned}\right\}\star$$

Case 1: When $\lambda=4$

$$-x+y=0$$
$$2x-2y=0$$
$$\Rightarrow \underline{\underline{x=y=1}}$$

$$\therefore X_1=\begin{bmatrix}1\\1\end{bmatrix}$$

Case 2: When $\lambda=1$

$$2x+y=0 \qquad 2x=-y$$
$$2x+y=0 \qquad \text{let } x=1,\ y=-2$$

$$\therefore X_2=\begin{bmatrix}1\\-2\end{bmatrix}$$

$$\therefore P=\begin{bmatrix}1&1\\1&-2\end{bmatrix}$$

Note that $P^{-1}AP=D$, where P & D are as given above.

$\therefore A$ is diagonalizable.

3. $\begin{bmatrix}-2&12\\-1&5\end{bmatrix}$

Characteristic eqn $|A-\lambda I|=0$

$$\begin{vmatrix}-2-\lambda&12\\-1&5-\lambda\end{vmatrix}=0$$

$$(-2-\lambda)(5-\lambda)+12=0$$
$$-10+2\lambda-5\lambda+\lambda^2+12=0$$
$$\lambda^2-3\lambda+2=0$$

$$\therefore \underline{\underline{\lambda=1,2}} \qquad \therefore D=\begin{bmatrix}1&0\\0&2\end{bmatrix}$$

Now, $[A-\lambda I][X]=0$

$$\left.\begin{aligned}(-2-\lambda)x+12y&=0\\ -x+(5-\lambda)y&=0\end{aligned}\right\}*$$

Case 1: $\lambda=1$

$-3x+12y=0$

$-x+4y=0$

$\underline{\underline{x=4y}}$ let $y=1$, $x=4$

$$X_1=\begin{bmatrix}4\\1\end{bmatrix}$$

Case 2: $\lambda=2$

$-4x+12y=0$

$-x+3y=0$

$\underline{x=3y}$ let $y=1$, $x=3$

$$X_2=\begin{bmatrix}3\\1\end{bmatrix}$$

$$\therefore P=\begin{bmatrix}4&3\\1&1\end{bmatrix}$$

Note that $P^{-1}AP=D$, where P & D are as given above. $\therefore A$ is diagonalizable.

4. $$\begin{bmatrix}1&6&1\\1&2&0\\0&0&3\end{bmatrix}$$

Characteristic eqn: $|A-\lambda I|=0$

$$\begin{vmatrix}1-\lambda&6&1\\1&2-\lambda&0\\0&0&3-\lambda\end{vmatrix}=0$$

$(1-\lambda)\left[(2-\lambda)(3-\lambda)-0\right]-6\left[(3-\lambda)\right]+1\left[0\right]=0$

$[3-\lambda]\left[(1-\lambda)(2-\lambda)-6\right]=0$

$(3-\lambda)\left[2-\lambda-2\lambda+\lambda^2-6\right]=0$

$(3-\lambda)\left[\lambda^2-3\lambda-4\right]=0$

$(3-\lambda)(\lambda+1)(\lambda-4)=0$

$\lambda = 3, -1, 4$

$$\therefore D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

Now, $[A - \lambda I][x] = [0]$

$(1-\lambda)x + 6y + z = 0$

$x + (2-\lambda)y = 0$

$(3-\lambda)z = 0$

Case 1: $\lambda = 3$

$-2x + 6y + z = 0$

$x - y = 0$ $\quad x = y$

$(0)z = 0$

Let $z = 1$

$-2x + 6x + 1 = 0$

$4x + 1 = 0$

$x = -1/4$

$= y$

$$X_1 = \begin{bmatrix} -1/4 \\ -1/4 \\ 1 \end{bmatrix}$$

Case 2: $\lambda = -1$

$* \Rightarrow 2x + 6y + z = 0$

$x + 3y = 0$ $\quad x = -3y$

$4z = 0$

$z = 0$

$2x - 6x + 0 = 0$

$\cancel{-4x = 0}$

$y = 1$

let $x = 3$,

$$X_2 = \begin{bmatrix} -3 \\ 1 \\ 0 \end{bmatrix}$$

Case 3: $\lambda = 4$

$*\Rightarrow$ Case 3: $\lambda = 4$

$-3x + 6y + z = 0$

$x - 2y = 0$

$-z = 0$

$\underline{z = 0}$

$x = 2y$

Let $y = 1$, $x = 2$

$\therefore X_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$

$\therefore P = \begin{bmatrix} -1/4 & -3 & 2 \\ -1/4 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

Note that $P^{-1}AP = D$ where P & D are given as above.

$\therefore$ A is diagonalised

5. $\begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$

Characteristic eqn, $|A - \lambda I| = 0$

$(1-\lambda)\left[(5-\lambda)(1-\lambda) - 1\right] - 1\left[(1-\lambda) - 3\right] + 3\left[1 - 3(5-\lambda)\right]$

$(1-\lambda)\left[5 - 5\lambda - \lambda + \lambda^2 - 1\right] - 1\left[-2 - \lambda\right] + 3\left[1 - 15 + 3\lambda\right]$

$(1-\lambda)\left[5 - 5\lambda - \lambda + \lambda^2 - 1\right] + 3\left[-14 + 3\lambda\right] - 1\left[-2 - \lambda\right]$

$(1-\lambda)\left[\lambda^2 - 6\lambda + 4)\right] \cancel{-39 + 6\lambda}$

$(1-\lambda)\left[\lambda^2 - 6\lambda + 4\right] - 42 + 9\lambda + 2 + \lambda$

$(1+\lambda)\left[\lambda^2 - 6\lambda + 4\right] - 40 + 10\lambda$

$\Rightarrow \lambda^3 - 7\lambda^2 + 36 = 0$

$\underline{\lambda = -2, 6, 3}$

$\therefore D = \begin{bmatrix} -2 & 0 & 0 \\ 0 & 6 & 0 \\ 0 & 0 & 3 \end{bmatrix}$

Now, $[A-\lambda I][x] = [0]$

$$\Rightarrow \begin{bmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = 0$$

$$\left.\begin{aligned} (1-\lambda)x + y + 3z &= 0 \\ x + (5-\lambda)y + z &= 0 \\ 3x + y + (1-\lambda)z &= 0 \end{aligned}\right\} *$$

$*\Rightarrow$ Case 1: $\lambda = -2$

$$\begin{aligned} 3x + y + 3z &= 0 \\ x + 7y + z &= 0 \\ 3x + y + 3z &= 0 \end{aligned}$$

$$\begin{array}{ccc} x & y & z \\ 3 & 1 & 3 \\ 1 & 7 & 1 \end{array}$$

$$\Rightarrow \frac{x}{\quad} = \frac{y}{\quad} = \frac{z}{\quad}$$

$$\begin{vmatrix} 1 & 3 \\ 7 & 1 \end{vmatrix} \quad \begin{vmatrix} 3 & 3 \\ 1 & 1 \end{vmatrix} \quad \begin{vmatrix} 3 & 1 \\ 1 & 7 \end{vmatrix}$$

$$\Rightarrow \frac{x}{-20} = \frac{y}{0} = \frac{z}{20}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -20 \\ 0 \\ 20 \end{bmatrix} = 20 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$X_1 = (-1, 0, 1)$$

$\lambda =$

$*\Rightarrow$ Case 2: $\lambda = 6$

$$\left.\begin{aligned} -5x + y + 3z &= 0 \\ x - y + z &= 0 \\ 3x + y - 5z &= 0 \end{aligned}\right\} *$$

$$\begin{array}{ccc} x & y & z \\ 1 & 3 & -5 \\ -1 & 1 & 1 \end{array} \begin{array}{c} \\ 1 \\ -1 \end{array}$$

$$= \frac{x}{4} = \frac{y}{8} = \frac{z}{4}$$

$$X_2 = \begin{bmatrix} 4 \\ 8 \\ 4 \end{bmatrix} = 4 \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} = (1 \ 2 \ 1)$$

Case 3: When $\lambda = 3$

$$* \Rightarrow \begin{aligned} -2x + y + 3z &= 0 \\ x + 2y + z &= 0 \\ 3x + y - 2z &= 0 \end{aligned}$$

$$\begin{array}{ccccc} x & & y & & z \\ 1 & \times & 3 & \times & -2 \\ 2 & \times & 1 & \times & 1 \end{array} \quad \begin{array}{c} 1 \\ 2 \end{array}$$

$$= \frac{x}{-5} = \frac{y}{5} = \frac{z}{-5}$$

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -5 \\ 5 \\ -5 \end{bmatrix} = 5 \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$$

$$X_3 = \begin{bmatrix} -1 \\ 1 \\ -1 \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} -1 & 1 & -1 \\ 0 & 2 & 1 \\ 1 & 1 & -1 \end{bmatrix}$$

QUADRATIC FORM:

A quadratic form on R^n is a function Q defined on R^n whose value at vector X in R^n can be computed by an expression $X^T A X$, where A is $n \times n$ symmetric matrix.

The matrix A is called matrix of the quadratic form.

Eg: Let $X = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. Compute $X^T A X$ for

$$A = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$$

$$X^T A X = \begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$= [x_1 \ x_2] \begin{bmatrix} 4x_1 \\ 3x_2 \end{bmatrix} = \underline{\underline{4x_1^2 + 3x_2^2}}$$

NOTE:

The matrix of the quadratic form is given by:

$$A = \begin{bmatrix} \text{coeff } x_1^2 & \frac{1}{2}\text{coeff } x_1x_2 & \cdots & \frac{1}{2}\text{coeff } x_1x_n \\ \frac{1}{2}\text{coeff } x_2x_1 & \text{coeff } x_2^2 & \cdots & \frac{1}{2}\text{coeff } x_2x_n \\ \vdots & & & \\ \frac{1}{2}\text{coeff } x_nx_1 & \frac{1}{2}\text{coeff } x_nx_2 & \cdots & \text{coeff } x_n^2 \end{bmatrix}$$

Eg: Let $q(x) = 5x_1^2 + 3x_2^2 + 2x_3^2 - x_1x_2 + 8x_2x_3$

$\Rightarrow$

$$A = \begin{bmatrix} 5 & -\frac{1}{2} & 0 \\ -\frac{1}{2} & 3 & 4 \\ 0 & 4 & 2 \end{bmatrix}_{3\times 3}$$

Change of variable in quadratic form:

If x represents a variable in R^n, then a change of variable is an equation of the form: $x = Py$ or $y = P^{-1}x$ —(1)

where P is an invertible matrix and y is a new variable.

If the change of variable (1) is made in a quadratic form, X^TAX then $X^TAX = (Py)^TA(Py)$

$$= y^T(P^TAP)y \quad \text{—(2)}$$

The new matrix of the quadratic form is P^TAP

If P orthogonally diagonalises A then $P^T = P^{-1}$ and

(2) $\Rightarrow$ $X^TAX = y^T(P^{-1}AP)y$

$= y^T D y$

Q) Reduce the following quadratic forms to canonical forms: (mentioning change of variable)

1. $x_1^2 - 8x_1x_2 - 5x_2^2$

$$A = \begin{bmatrix} 1 & -4 \\ -4 & -5 \end{bmatrix}_{2\times 2}$$

Characteristic equation, $|A - \lambda I| = 0$

$(1-\lambda)(-5-\lambda) - 16 = 0$

$-5 - \lambda + 5\lambda + \lambda^2 - 16 = 0$

$\lambda^2 + 4\lambda - 21 = 0$

$\underline{\underline{\lambda = 3, -7}}$

$$\therefore D = \begin{bmatrix} 3 & 0 \\ 0 & -7 \end{bmatrix} \Rightarrow \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix}$$

Now, $[A - \lambda I][x] = [0]$

$$= \begin{bmatrix} 1-\lambda & -4 \\ -4 & -5-\lambda \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\left.\begin{aligned} (1-\lambda)x - 4y &= 0 \\ -4x + (-5-\lambda)y &= 0 \end{aligned}\right\}^*$$

Let change of variable $x = py$

To find p:

Case 1: $\lambda = 3$

$-2x - 4y = 0 \quad \Rightarrow \quad -2x = 4y$

$-4x - 8y = 0 \qquad\qquad -x = 2y$

Let $y = 1$, $x = -2$

$$X_1 = \begin{bmatrix} -2 \\ 1 \end{bmatrix} \quad P_2 = \begin{bmatrix} -2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$= \begin{bmatrix} -2/\sqrt{1^2+(-2)^2} \\ 1/\sqrt{(-2)^2+(1)^2} \end{bmatrix}$$

case 2: $\lambda = -7$

$8x - 4y = 0$

$-4x + 2y = 0$

$\Rightarrow 8x = 4y$

$\underline{2x = y}$ let $x = 1, y = 2$

$$X_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$ [normalising, i.e. $\frac{1}{\sqrt{1^2+2^2}}$]

$$P_1 = \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix}$$

We have, $x = py$

$$\therefore x^T A x = (py)^T A py$$
$$= y^T (P^T A P) y$$
$$= y^T D y$$
$$= [y_1 \ y_2] \begin{bmatrix} -7 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= [y_1 \ y_2] \begin{bmatrix} -7y_1 \\ 3y_2 \end{bmatrix}$$

$$= x^T A x$$
$$= -7y_1^2 + 3y_2^2$$ is canonical form & change of variable $x = py$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{5} & -2/\sqrt{5} \\ 2/\sqrt{5} & 1/\sqrt{5} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\Rightarrow x_1 = \frac{1}{\sqrt{5}} y_1 - \frac{2}{\sqrt{5}} y_2$$

$$x_2 = \frac{2}{\sqrt{5}} y_1 + \frac{1}{\sqrt{5}} y_2$$

2. $3x_1^2 - 4x_1x_2 + 6x_2^2$

$$A = \begin{bmatrix} 3 & -2 \\ -2 & 6 \end{bmatrix}$$

Characteristic equation, $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & -2 \\ -2 & 6-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)(6-\lambda) - 4 = 0$$

$$18 - 3\lambda - 6\lambda + \lambda^2 - 4 = 0$$

$$\lambda^2 - 9\lambda + 14 = 0$$

$$\lambda = 7, 2$$

$$\therefore D = \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix}$$

Now, $[A - \lambda I][X] = [0]$ → Let change of variable $x = py$.

$$\left.\begin{aligned} (3-\lambda)x - 2y &= 0 \\ -2x + (6-\lambda)y &= 0 \end{aligned}\right\} *$$

To find p:

Case 1: $\lambda = 2$

$x - 2y = 0$ $\Rightarrow x = 2y$

$-2x + 4y = 0$ Let $y = 1$, $x = 2$

$$X_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \qquad p_1 = \begin{bmatrix} 2/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

Case 2: $\lambda = 7$

$-4x - 2y = 0$

$-2x - y = 0$

$\Rightarrow -2x = y$

let $x = 1, y = -2$

$X_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \Rightarrow P_2 = \begin{bmatrix} 1/\sqrt{5} \\ -2/\sqrt{5} \end{bmatrix}$

$\therefore P = \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & -2/\sqrt{5} \end{bmatrix}$

We have $x = py$

$\therefore x^T A x = (py)^T A (py)$

$= y^T P^T A P y$

$= y^T D y$

$= [y_1 \quad y_2] \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$= [y_1 \quad y_2] \begin{bmatrix} 2y_1 \\ 7y_2 \end{bmatrix}$

$x^T A x = 2y_1^2 + 7y_2^2$ is canonical form

& change of variable

$x = py \Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ 1/\sqrt{5} & -2/\sqrt{5} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

$\therefore x_1 = \frac{2}{\sqrt{5}} y_1 + \frac{1}{\sqrt{5}} y_2$

$x_2 = \frac{1}{\sqrt{5}} y_1 - \frac{2}{\sqrt{5}} y_2$

3. $x_1^2 - 6x_1x_2 + 9x_2^2$

$$A = \begin{bmatrix} 1 & -3 \\ -3 & 9 \end{bmatrix}$$

Characteristic eqn : $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & -3 \\ -3 & 9-\lambda \end{vmatrix} = 0$$

$$(1-\lambda)(9-\lambda) - 9 = 0$$

$$9 - \lambda - 9\lambda + \lambda^2 - 9 = 0$$

$$\lambda^2 - 10\lambda = 0$$

$$\underline{\underline{\lambda = 10, 0}}$$

$$\therefore D = \begin{bmatrix} 0 & 0 \\ 0 & 10 \end{bmatrix}$$

Now, $[A - \lambda I][x] = [0]$

$$\left.\begin{aligned} (1-\lambda)x - 3y &= 0 \\ -3x + (9-\lambda)y &= 0 \end{aligned}\right\} *$$

Let change of variable $x = py$

To find p:

Case 1: $\lambda = 0$

$$x - 3y = 0$$

$$-3x + 9y = 0$$

$$\Rightarrow x = 3y$$

Let $y = 1$, $x = 3$

$$\therefore X_1 = \begin{bmatrix} 3 \\ 1 \end{bmatrix}, \quad p_1 = \begin{bmatrix} 3/\sqrt{10} \\ 1/\sqrt{10} \end{bmatrix}$$

Case 2: $\lambda = 10$

$$-9x - 3y = 0$$

$$-3x - y = 0$$

$$\Rightarrow -3x = y$$

Let $x = 1$, $y = -3$

$$\therefore X_2 = \begin{bmatrix} 1 \\ -3 \end{bmatrix}, \quad p_2 = \begin{bmatrix} 1/\sqrt{10} \\ -3/\sqrt{10} \end{bmatrix}$$

$$\therefore P = \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix}$$

$$\Rightarrow x^T A x = y^T D y$$

$$= [y_1 \;\; y_2] \begin{bmatrix} 0 & 0 \\ 0 & 10 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= [y_1 \;\; y_2] \begin{bmatrix} 0 \\ 10y_2 \end{bmatrix}$$

$$= \underline{\underline{0y_1 + 10y_2^2}} \text{ is canonical form}$$

& change of variable $x = Py$

$$\Rightarrow \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 3/\sqrt{10} & 1/\sqrt{10} \\ 1/\sqrt{10} & -3/\sqrt{10} \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$\therefore x_1 = \frac{3}{\sqrt{10}} y_1 + \frac{1}{\sqrt{10}} y_2$$

$$x_2 = \frac{1}{\sqrt{10}} y_1 - \frac{3}{\sqrt{10}} y_2$$

ASSIGNMENT:

4. $2x_1^2 + 10x_1x_2 + 2x_2^2$

CLASSIFICATION OF QUADRATIC FORMS:

The quadratic form Q is (a) positive definite if $Q(x) > 0$ for all $x \neq 0$.
(b) negative definite if $Q(x) < 0$ for all $x \neq 0$
(c) indefinite if $Q(x)$ takes both positive and negative values.

→ Quadratic forms and eigen values –
Let A be $n \times n$ symmetric matrix, then quadratic form $x^T A x$ is : (a) positive definite if & only if the eigen values of A are all positive.
(b) negative definite if & only if the eigen values of A are all negative
(c) indefinite if & only if A has both +ve & -ve values

1. Check whether following quadratic forms are positive, negative definite or indefinite.

i) $Q(x) = 3x_1^2 + 2x_2^2 + x_3^2 + 4x_1x_2 + 4x_2x_3$

$$\therefore A = \begin{bmatrix} 3 & 2 & 0 \\ 2 & 2 & 2 \\ 0 & 2 & 1 \end{bmatrix}$$

Characteristic equation, $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & 2 & 0 \\ 2 & 2-\lambda & 2 \\ 0 & 2 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (3-\lambda)\left[(2-\lambda)(1-\lambda) - 4\right] - 2\left[(2)(1-\lambda) - 0\right] + 0 = 0$$

$$(3-\lambda)\left[2 - 2\lambda - \lambda + \lambda^2 - 4\right] - 2\left[2 - 2\lambda\right]$$

$$= (3-\lambda)\left[\lambda^2 - 3\lambda - 2\right] - 4\left[1 - \lambda\right]$$

$$= 3\lambda^2 - 9\lambda - 6 - \lambda^3 + 3\lambda^2 + 2\lambda - 4 + 4\lambda$$

$$= -\lambda^3 + 6\lambda^2 - 3\lambda - 10 = 0$$

$$= \lambda^3 - 6\lambda^2 + 3\lambda + 10 = 0$$

$$\underline{\underline{\lambda = -1, 5, 2}}$$

$\therefore Q(x)$ is indefinite.

2. $3x_1^2 + 5x_2^2 + 3x_3^2 - 2x_1x_2 + 2x_1x_3 - 2x_2x_3$

$$\therefore A = \begin{bmatrix} 3 & -1 & 1 \\ -1 & 5 & -1 \\ 1 & -1 & 3 \end{bmatrix}$$

characteristic equation, $|A - \lambda I| = 0$

$$\begin{vmatrix} 3-\lambda & -1 & 1 \\ -1 & 5-\lambda & -1 \\ 1 & -1 & 3-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)\left[(5-\lambda)(3-\lambda) - 1\right] + 1\left[-1(3-\lambda) + 1\right] + 1\left[1 - 1(5-\lambda)\right]$$

$$(3-\lambda)\left[15 - 5\lambda - 3\lambda + \lambda^2 - 1\right] + \left[-3 + \lambda + 1\right] + \left[1 - 5 + \lambda\right]$$

$$(3-\lambda)\left[14 - 8\lambda + \lambda^2\right] + \left[\lambda - 2\right] + \left[\lambda - 4\right]$$

$$= 3\lambda^2 - 24\lambda + 42 - \lambda^3 + 8\lambda^2 - 14\lambda + \lambda - 2 + \lambda - 4$$

$$= -\lambda^3 + 11\lambda^2 - 36\lambda + 36$$

$\underline{\lambda = 6, 3, 2}$

$\therefore$ $Q(x)$ is positive definite.

<u>QR Factorization:</u>

If A is $M \times N$ matrix with linearly independent columns, then A can be factored as $A = QR$, where Q is $m \times n$ matrix whose columns are orthonormal basis for column space of A. & R is $n \times n$ upper triangular invertible matrix with positive entries on its diagonal.

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→ Obtain the QR factorization of the matrix A:

1) $A = \begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$

$x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $x_2 = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$

$v_1 = x_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$, $\|v_1\| = \sqrt{v_1 \cdot v_1} = \underline{\underline{\sqrt{2}}}$

$\rightarrow x_2 \cdot v_1 = x_2^T v_1 = \begin{bmatrix} -1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \underline{\underline{-1}}$

$\Rightarrow v_1 \cdot v_1 = \begin{bmatrix} 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = 1+1 = \underline{\underline{2}}$

$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} \cdot v_1$

$= \begin{bmatrix} -1 \\ 0 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

$= \underline{\underline{\begin{bmatrix} -1/2 \\ -1/2 \end{bmatrix}}}$

$\|v_2\| = \sqrt{v_2 \cdot v_2}$

$= \sqrt{(-1/2)^2 + (-1/2)^2} = \sqrt{\frac{1}{4} + \frac{1}{4}} = \underline{\underline{\frac{1}{\sqrt{2}}}}$

$\therefore Q = \begin{bmatrix} \frac{v_1}{\|v_1\|}, & \frac{v_2}{\|v_2\|} \end{bmatrix}$

$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$

$R = Q^T A$

$= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1 & -1 \\ -1 & 0 \end{bmatrix}$

$= \begin{bmatrix} 1/\sqrt{2} + 1/\sqrt{2} & -1/\sqrt{2} \\ -1/\sqrt{2} + 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} = \underline{\underline{\begin{bmatrix} \sqrt{2} & -1/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{bmatrix}}}$

2) $A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

$x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad x_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$v_1 = x_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \|v_1\| = \sqrt{v_1 \cdot v_1} = \sqrt{4+1} = \underline{\underline{\sqrt{5}}}$

$v_2 = x_2 - \dfrac{x_2 \cdot v_1}{v_1 \cdot v_1} \cdot v_1$

$\rightarrow x_2 \cdot v_1 = x_2^T v_1 = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \underline{\underline{3}}$

$= \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \dfrac{3}{5} \cdot \begin{bmatrix} 2 \\ 1 \end{bmatrix}$

$\rightarrow v_1 \cdot v_1 = v_1^T v_1 = \begin{bmatrix} 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \underline{\underline{5}}$

$v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 6/5 \\ 3/5 \end{bmatrix}$

$v_2 = \underline{\underline{\begin{bmatrix} -1/5 \\ 2/5 \end{bmatrix}}}$

$\|v_2\| = \sqrt{v_2 \cdot v_2} = \sqrt{\left(\frac{-1}{5}\right)^2 + \left(\frac{2}{5}\right)^2}$

$= \sqrt{\frac{1}{25} + \frac{4}{25}} = \sqrt{\frac{5}{25}}$

$= \underline{\underline{\frac{1}{\sqrt{5}}}}$

$Q = \begin{bmatrix} \frac{v_1}{\|v_1\|} & \frac{v_2}{\|v_2\|} \end{bmatrix} = \begin{bmatrix} 2/\sqrt{5} & -1/\sqrt{5} \\ 1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix}$

$R = Q^T A = \begin{bmatrix} 2/\sqrt{5} & 1/\sqrt{5} \\ -1/\sqrt{5} & 2/\sqrt{5} \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}$

$$= \begin{bmatrix} \frac{4}{\sqrt{5}} + \frac{1}{\sqrt{5}} & \frac{2}{\sqrt{5}} + \frac{1}{\sqrt{5}} \\ \frac{-2}{\sqrt{5}} + \frac{2}{\sqrt{5}} & \frac{-1}{\sqrt{5}} + \frac{2}{\sqrt{5}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{5}{\sqrt{5}} & \frac{3}{\sqrt{5}} \\ 0 & \frac{1}{\sqrt{5}} \end{bmatrix} = \begin{bmatrix} \sqrt{5} & \frac{3}{\sqrt{5}} \\ 0 & \frac{1}{\sqrt{5}} \end{bmatrix}$$

3) $A = \begin{bmatrix} 3 & 8 \\ 0 & 5 \\ -1 & -6 \end{bmatrix}$

$a_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$ $\quad a_2 = \begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix}$

$V_1 = a_1 = \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$, $\|V_1\| = \sqrt{9+1} = \sqrt{10}$

$V_2 = a_2 - \frac{a_2 \cdot V_1}{V_1 \cdot V_1} \cdot V_1$

$\rightarrow a_2 \cdot V_1 = a_2^T V_1 = \begin{bmatrix} 8 & 5 & -6 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = 30$

$\rightarrow V_1 \cdot V_1 = V_1^T V_1 = \begin{bmatrix} 3 & 0 & -1 \end{bmatrix} \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix} = 10$

$= \begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix} - \frac{30}{10} \begin{bmatrix} 3 \\ 0 \\ -1 \end{bmatrix}$

$= \begin{bmatrix} 8 \\ 5 \\ -6 \end{bmatrix} - \begin{bmatrix} 9 \\ 0 \\ -3 \end{bmatrix}$

$= \begin{bmatrix} -1 \\ 5 \\ -3 \end{bmatrix}$

$\|V_2\| = \sqrt{1+25+9}$

$= \sqrt{35}$

$$Q = \begin{bmatrix} \frac{v_1}{\|v_1\|} & \frac{v_2}{\|v_2\|} \end{bmatrix}$$

$$= \begin{bmatrix} 3/\sqrt{10} & -1/\sqrt{35} \\ 0 & 5/\sqrt{35} \\ -1/\sqrt{10} & -3/\sqrt{35} \end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix} \frac{3}{\sqrt{10}} & 0 & -1/\sqrt{10} \\ -1/\sqrt{35} & 5/\sqrt{35} & -3/\sqrt{35} \end{bmatrix} \begin{bmatrix} 3 & 8 \\ 0 & 5 \\ -1 & -6 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{9}{\sqrt{10}} + \frac{1}{\sqrt{10}} & \frac{24}{\sqrt{10}} + \frac{6}{\sqrt{10}} \\ \frac{-3}{\sqrt{35}} + \frac{3}{\sqrt{35}} & \frac{-8}{\sqrt{35}} + \frac{25}{\sqrt{35}} + \frac{18}{\sqrt{35}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{10}{\sqrt{10}} & \frac{30}{\sqrt{10}} \\ 0 & \frac{35}{\sqrt{35}} \end{bmatrix} = \begin{bmatrix} \sqrt{10} & 3\sqrt{10} \\ 0 & \sqrt{35} \end{bmatrix}$$

4) $\begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

$a_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ $\quad a_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ $\quad a_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$v_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

$\|v_1\| = \sqrt{1^2 + 1^2} = \sqrt{2}$

$$v_2 = x_2 - \frac{x_2 \cdot v_1}{v_1 \cdot v_1} \cdot v_1$$

$$x_2^T v_1 = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \underline{1}$$

$$v_1^T v_1 = \begin{bmatrix} 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \underline{\underline{2}}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

$$v_2 = \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}$$

$$\Rightarrow \|v_2\| = \sqrt{(1/2)^2 + (-1/2)^2 + (1)^2}$$

$$= \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \underline{\underline{\sqrt{\frac{6}{4}}}} = \underline{\underline{\sqrt{\frac{3}{2}}}}$$

$$v_3 = x_3 - \frac{x_3 \cdot v_1}{v_1 \cdot v_1} \cdot v_1 - \frac{x_3 \cdot v_2}{v_2 \cdot v_2} \cdot v_2$$

$$x_3 \cdot v_1 = x_3^T v_1 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \underline{\underline{1}}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} - \frac{1/2}{3/2} \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}$$

$$v_1 \cdot v_1 = \underline{\underline{2}}$$

$$= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} 1/2 \\ 1/2 \\ 0 \end{bmatrix} - \frac{1}{3} \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}$$

$$x_3 \cdot v_2 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}$$

$$= \underline{\underline{1/2}}$$

$$v_2 \cdot v_2 = \begin{bmatrix} 1/2 & -1/2 & 1 \end{bmatrix} \begin{bmatrix} 1/2 \\ -1/2 \\ 1 \end{bmatrix}$$

$$= \frac{1}{4} + \frac{1}{4} + 1$$

$$= \underline{\underline{\frac{3}{2}}}$$

$$v_3 = \begin{bmatrix} -2/3 \\ 2/3 \\ 2/3 \end{bmatrix}$$

$$\|v_3\| = \sqrt{(2/3)^2 + (2/3)^2 + (2/3)^2} = \sqrt{\frac{4}{9} + \frac{4}{9} + \frac{4}{9}} = \sqrt{\frac{\cancel{12}\,4}{\cancel{9}\,3}}$$

$$= \underline{\underline{2/\sqrt{3}}}$$

$$\therefore Q = \begin{bmatrix} \frac{V_1}{\|V_1\|} & \frac{V_2}{\|V_2\|} & \frac{V_3}{\|V_3\|} \end{bmatrix}$$

$$\frac{1}{2} \times \frac{\sqrt{2}}{\sqrt{3}}$$

$$\frac{1}{\sqrt{2}\,\cancel{\sqrt{2}}} \times \frac{\cancel{\sqrt{2}}}{\sqrt{3}}$$

$$= \frac{1}{\sqrt{6}}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1/2}{\sqrt{3}/2} & -2/3 \times \frac{\sqrt{3}}{2} \\ \frac{1}{\sqrt{2}} & \frac{-1/2}{\sqrt{3}/2} & \frac{2}{3} \times \frac{\sqrt{3}}{2} \\ 0 & \frac{1}{\sqrt{3}/2} & \frac{2}{3} \times \frac{\sqrt{3}}{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -1/\sqrt{6} & \frac{1}{\sqrt{3}} \\ 0 & \frac{\sqrt{2}}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$R = Q^T A$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & \frac{\sqrt{2}}{\sqrt{3}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

$$\frac{1}{\sqrt{6}} + \frac{\sqrt{2}}{\sqrt{3}}$$

$$= \frac{1}{\sqrt{6}} + \frac{2}{}$$

$$= \frac{3}{\sqrt{6}}$$

$$= \frac{\sqrt{3}\sqrt{3}}{\sqrt{2}\sqrt{3}}$$

$$= \begin{bmatrix} \frac{2}{\sqrt{2}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{\frac{3}{2}} & \frac{1}{\sqrt{6}} \\ 0 & 0 & \frac{2}{\sqrt{3}} \end{bmatrix}$$

5] $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

$x_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ $x_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$ $x_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$

$v_1 = x_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ $\|v_1\| = \underline{\underline{\sqrt{2}}}$

$\Rightarrow x_2 \cdot v_1 = \begin{bmatrix} 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \underline{\underline{2}}$

$v_2 = x_2 - \dfrac{x_2 \cdot v_1}{v_1 \cdot v_1} \cdot v_1$

$= \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \dfrac{2}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$v_1 \cdot v_1 = \begin{bmatrix} 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \underline{\underline{2}}$

$v_2 = \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$ $\|v_2\| = \underline{\underline{\sqrt{3}}}$

$v_3 = x_3 - \dfrac{x_3 \cdot v_1}{v_1 \cdot v_1} \cdot v_1 - \dfrac{x_3 \cdot v_2}{v_2 \cdot v_2} \cdot v_2$

$= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} - \dfrac{1}{2} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} - 0 \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$

$\Rightarrow x_3 \cdot v_1 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \underline{\underline{1}}$

$v_1 \cdot v_1 = \underline{\underline{2}}$

$x_3 \cdot v_2 = \begin{bmatrix} 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} = \underline{\underline{0}}$

$= \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} 1/2 \\ 0 \\ 1/2 \end{bmatrix}$

$v_3 = \begin{bmatrix} -1/2 \\ 1 \\ 1/2 \end{bmatrix}$ $\|v_3\| = \sqrt{(1/2)^2 + (1/2)^2 + (1)^2} = \sqrt{\frac{1}{4} + \frac{1}{4} + 1} = \underline{\underline{\frac{\sqrt{3}}{\sqrt{2}}}}$

$$\therefore Q = \begin{bmatrix} \frac{v_1}{\|v_1\|} & \frac{v_2}{\|v_2\|} & \frac{v_3}{\|v_3\|} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{2} \times \frac{\sqrt{2}}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{3}} & 1 \times \frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{2} \times \frac{\sqrt{2}}{\sqrt{3}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{\sqrt{2}}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix}$$

$R = Q^T A$

$$= \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ -\frac{1}{\sqrt{6}} & \frac{\sqrt{2}}{\sqrt{3}} & \frac{1}{\sqrt{6}} \end{bmatrix} \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \frac{3}{\sqrt{3}} & 0 \\ 0 & -\frac{2}{\sqrt{6}} + \sqrt{\frac{2}{3}} & \sqrt{\frac{2}{3}} + \frac{1}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} \sqrt{2} & \sqrt{2} & \frac{1}{\sqrt{2}} \\ 0 & \sqrt{3} & 0 \\ 0 & \frac{-2+2\sqrt{2}}{\sqrt{6}} & \frac{3}{\sqrt{6}} \end{bmatrix}$$

$$-\frac{2}{\sqrt{6}} + \frac{2\sqrt{2}}{\sqrt{6}} \Rightarrow \frac{\sqrt{2}+1}{\sqrt{3}\ \sqrt{6}}$$

$$= \frac{2}{\sqrt{6}} + \frac{1}{\sqrt{6}} =$$

ASSIGNMENT:

1) $\begin{bmatrix} 0 & 1 \\ -1 & 0 \\ 1 & -1 \end{bmatrix}$ 2) $\begin{bmatrix} 3 & 1 \\ 6 & 2 \\ 0 & 2 \end{bmatrix}$

LEAST SQUARE METHOD:

Used to find the solution of inconsistent system $AX=B$, when a solution doesn't exist, the best one can do is to find X that makes AX as close as possible to B. The general least square method is to find X that makes $\|B-AX\|$ as small as possible.

To find Least square solution of $AX=B$:

(1) The least square solution of equation $AX=B$ satisfies the equation $A^TAX = A^TB$

(2) If the matrix A^TA is invertible, the equation $AX=B$ has only one least square solution. $\hat{X}$ and is given by $\hat{X} = (A^TA)^{-1} \cdot A^TB$

1. Find a least square solution of the inconsistent system, $AX=B$ for

$A = \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix}$, $B = \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix}$

ANS- Least square solution of $AX=B$ satisfies the equation $A^TAX = A^TB$ & solution is $\hat{X} = (A^TA)^{-1} \cdot A^TB$ — (1)

$$A^TA = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 2 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 17 & 1 \\ 1 & 5 \end{bmatrix}$$

$$(A^TA)^{-1} = \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix}$$

$$A^TB = \begin{bmatrix} 4 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 11 \end{bmatrix} = \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$① \Rightarrow \hat{X} = \frac{1}{84} \begin{bmatrix} 5 & -1 \\ -1 & 17 \end{bmatrix} \begin{bmatrix} 19 \\ 11 \end{bmatrix}$$

$$= \frac{1}{84} \begin{bmatrix} 84 \\ 168 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$\therefore$ solution is $\hat{X} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

2. Find a least square solution of $AX = B$ for

$$A = \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}, \quad B = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

ANS- Least square solution of $AX = B$ satisfies the equation $A^TAX = A^TB$ & solution is

$$\hat{X} = (A^TA)^{-1}A^TB \quad — ①$$

$$A^TA = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} -1 & 2 \\ 2 & -3 \\ -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 6 & -11 \\ -11 & 22 \end{bmatrix}$$

$$(A^TA)^{-1} = \frac{1}{11} \begin{bmatrix} 22 & 11 \\ 11 & 6 \end{bmatrix}$$

$$A^TB = \begin{bmatrix} -1 & 2 & -1 \\ 2 & -3 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

$$① \Rightarrow \hat{X} = \frac{1}{11} \begin{bmatrix} 22 & 11 \\ 11 & 6 \end{bmatrix} \begin{bmatrix} -4 \\ 11 \end{bmatrix}$$

$$= \frac{1}{11} \begin{bmatrix} 33 \\ 22 \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$\therefore$ Solution is $\hat{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

3. Find least square solution of $AX = B$ for
$A = \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix}$, $B = \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix}$

ANS- least square soln. of $AX = B$ satisfies the equation $A^TAX = A^TB$ & solution is $\hat{x} = (A^TA)^{-1}A^TB$ — (1)

$$A^TA = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ -2 & 0 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 12 & 8 \\ 8 & 10 \end{bmatrix}$$

$$(A^TA)^{-1} = \frac{1}{56} \begin{bmatrix} 10 & -8 \\ -8 & 12 \end{bmatrix}$$

$$A^TB = \begin{bmatrix} 2 & -2 & 2 \\ 1 & 0 & 3 \end{bmatrix} \begin{bmatrix} -5 \\ 8 \\ 1 \end{bmatrix} = \begin{bmatrix} -24 \\ -2 \end{bmatrix}$$

$$(1) \Rightarrow \hat{x} = \frac{1}{56} \begin{bmatrix} 10 & -8 \\ -8 & 12 \end{bmatrix} \begin{bmatrix} -24 \\ -2 \end{bmatrix} = \frac{1}{56} \begin{bmatrix} -224 \\ 168 \end{bmatrix}$$

$\therefore$ $\hat{x} = \begin{bmatrix} -4 \\ 3 \end{bmatrix}$ is solution